

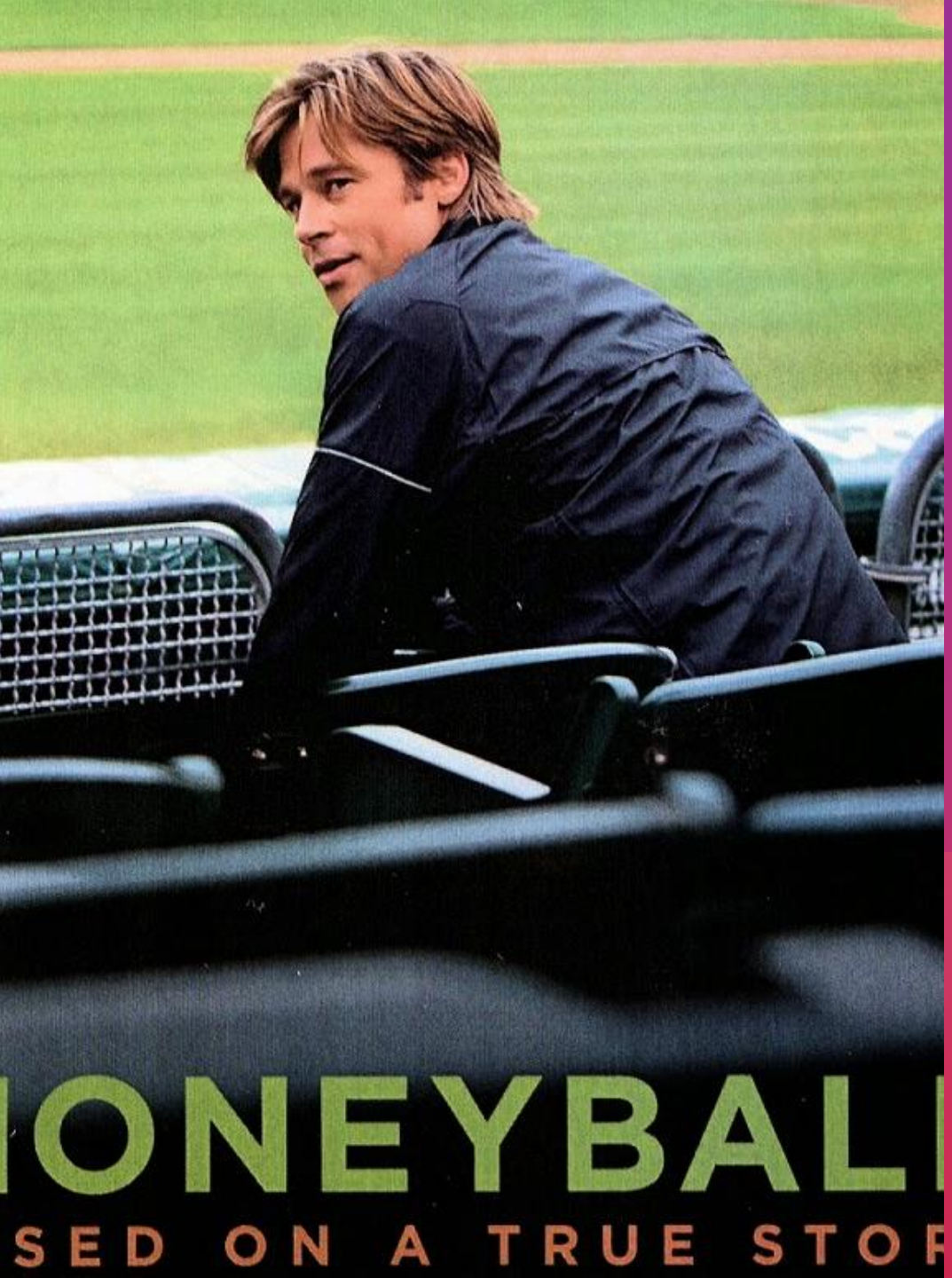
BASEBALL AND STATISTICS

SESSION 2-ADVANCED METRICS

Review of Session 1

- History of Statistics in Baseball and Evolution of the Game
  - Batting Average, HRs
  - Wins, ERA
  - Fielding %
- Performance of Twins Players
  - Season and Career
- Advanced Metrics
  - Sabermetrics
  - MoneyBall





EVOLUTION OF ADVANCED METRICS

Bill James and Sabermetrics

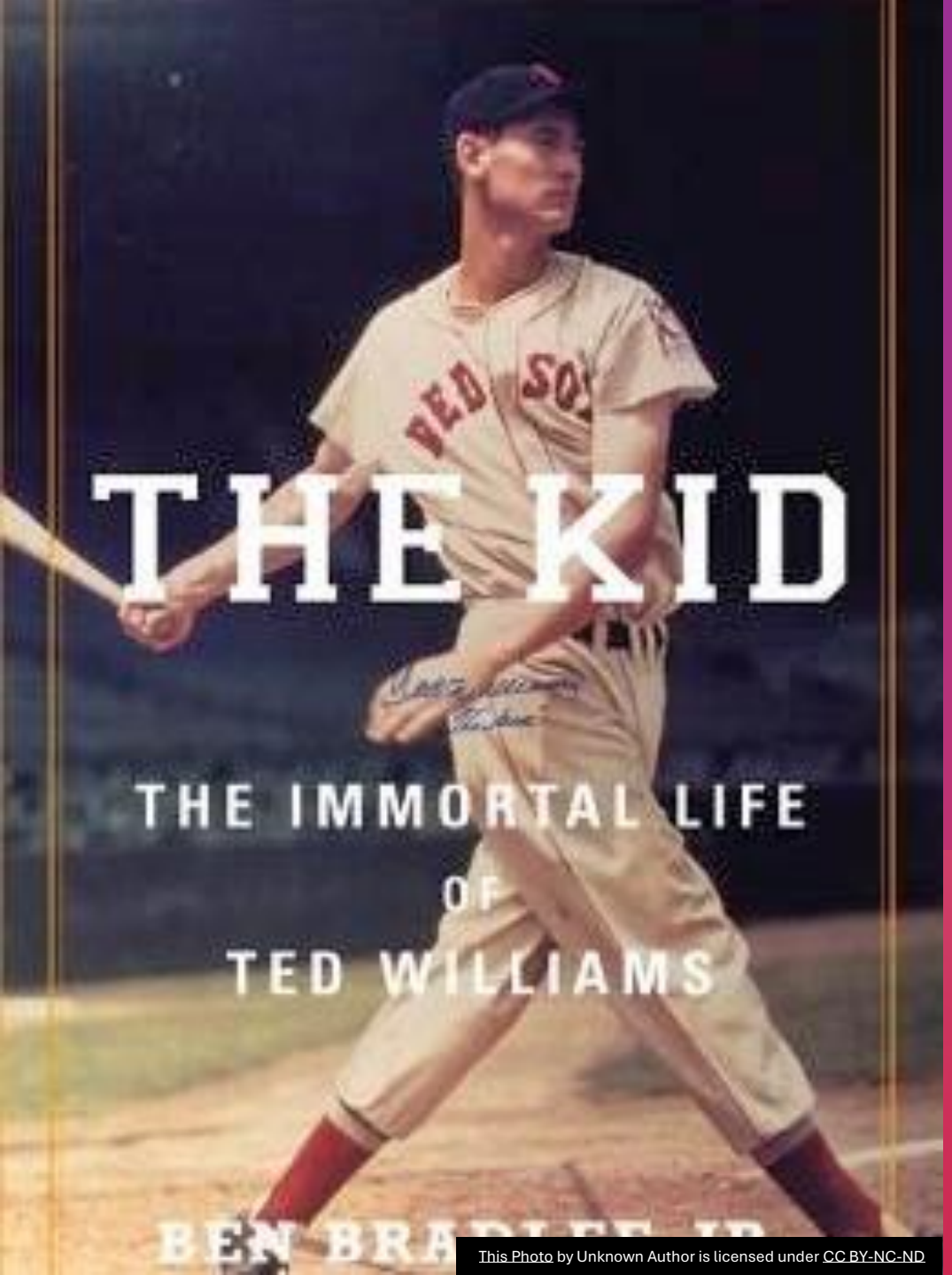
- Often considered the “father of advanced baseball analytics”
- Defined as “the search for objective knowledge about baseball”
- His analysis and writings about baseball data analytics started to get traction in the 1980’s
  - Introduced ideas such as WAR (Wins Above Replacment)
- Named to the list of 100 most influential people by Time in early 2000’s
- Hired by the Red Sox in 2003 and they subsequently won 2 World Series
- Concepts carried on the Society for American Baseball Research (SABR)

Moneyball: The Art of Winning an Unfair Game

- Book by Michael Lewis on General Manager Billy Beane's use of Data Analytics to transform the Oakland A's
- In 2002 the A's payroll was more than $20M less then rival teams
- Based upon his experience as a prospect for the NY Mets, Beane thought that current player scouting methods were inadequate
    - Instincts versus data
- Focused instead on Sabermetric approaches
    - On-base percentage + Slugging percentage to evaluate hitters
- Results
    - Made the playoffs in 2002, with record 20 game win steak
    - Made the playoffs 8 times during his tenure
        - Won only one playoff series (2006 vs Minnesota Twins
        - Never played in league championship series or won a World Series
        - Approach adopted by many teams and now all teams have a "Data Analytics Group in the Front Office helping guide strategy and in-game decision making



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SESSION 2

ADVANCED METRICS

On Base Percentage-A Walk is as Good as a Hit

Ways to get on Base

- Hit
- Walk
- Hit by Pitch
- Error
- Fielder's Choice
- Dropped 3rd Strike
- Catcher's Interference
- Obstruction

OBP = Hits + Walks+ Hit By Pitch/Total At Bats (Hits, Walks, HBP, Sac, Sac Flys)

Rating	OBP
Excellent	.390 and above
Great	.370
Above Average	.340
Average	.320
Below Average	.310
Poor	.300
Awful	.290 or below

OBP Examples

- 1941 Joe Dimaggio vs Ted Williams
  - Season of Joe D 56 game hitting streak and Ted W .406 batting average
  - Joe D OBP .440 and Ted W .553
  - Joe D AL MVP
- Joe Mauer and HOF Catchers
  - Ranks 3rd (.388) only behind Josh Gibson (.458), Mickey Cochran (.419)

Player-MLB	OBP
Y. Alvarez	.432
TJ Rumfield	.390
J. Bauers	.389
J. Wood	.388
G. Moreno	.388
Player-Twins	OBP
L. Keaschall	.374
R. Jeffers	.353
T. Larnach	.352
A. Martin	,341
J. Bell and B. Buxton	,319

MLB Avg OBP .319

OPS-On Base Percentage + Slugging Percentage

- Combines the players ability to get on base with ability to hit with power
- Example
  - Roberto Clemente in 1967
  - OBP .400 and SLG .574
  - OPS .974
- Babe Ruth
  - Led the AL in HRs 12 times, but led the AL in OPS 13 times

Rating	OPS
Excellent	1.000
Great	.900
Above Average	.800
Average	.700
Below Average	.670
Poor	.600
Awful	.570

Batting statistics are now often represented by what Is called the slash line
BA/OBP/OPS

OPS Leaders-2026

MLB

Player	OPS
Y. Alvarez	1.033
P. Crow-Armstrong	.947
W. Contreras	.901
B. Rice	.897
J. Wood	.891

Twins

Player	OPS
B. Buxton	.829
R. Jeffer	.814
T. Larnach	.788
L. Keaschall	.781
C. Clemens	.758

MLB OPS Avg .719

Runs Created (RC)

- An estimate of the player's total offensive contribution to his team's total runs
- A simple way to evaluate is Runs Scored+ Runs Batted In-HRs
  - H. Killebrew 1964 (95+111-49)= 157
- The Sabermetric calculation (Bill James)
  - RC = (Hits+Walks+HBP-Caught Stealing-Grounded into DP) x (Total Bases+0.26(Walks-Int Walks+HBP) + (.52 (Sacrifice Bunts+Sacrifice Flys+Stolen Bases)/At Bats+Walks+HBP+Sac Bunts+Sac Flys)
    - The .26 and .52 weights were created by Bill James to give walks, hbp, sacrifices their relation to runs scored
  - Examples
    - 2026 leaders-Y. Alvarez 155 and P. Crow-Armstrong 145

Other Metrics

- Exit Velocity
  - How fast in mph the ball comes off the bat
- Launch Angle
  - The vertical angle in degrees at which the ball is hit in the air
- Hard Hit Rate
  - % of balls with >95 mph exit velocity
- Barrel Rate
  - Combines exit velocity and launch angle
- Others
  - Chase rate, Whiff rate, Isolated Power



ADVANCED PITCHING METRICS

WHIP (Walks + Hits/Inning Pitched)

- A measure of how well the pitcher kept runners off the basepath
- Example
  - Tom Seaver 1971 NY Mets
  - 61 walks, 210 hits allowed in 2861/3 IP
  - 61+210/286.3 = WHIP .946
- Twins best single season WHIP
  - Johan Santana 0.92

Rating	WHIP
Excellent	1.00 and under
Great	1.10
Above Average	1.20
Average	1.30
Below Average	1.40
Poor	1.50
Awful	1.60 and higher

Top Single Season WHIP is Pedro Martinez in 2002
WHIP 0.737 with league average of 1.414
2026 leader is Jacob Misiorowski 0.803 with league Average of 1.305

WHIP Leaders-2026

MLB

Player	WHIP
J. Misiorowski	0.80
Y. Yamamoto	0.87
D. Rasmussen	0.91
C. Schlittler	0.92
C. Sale	0.98

Twins

Player	WHIP
J. Ryan	1.16
T. Bradley	1.22
Z. Matthews	1.26
C. Prielipp	1.29
B. Ober	1.30

ERA+

- A pitcher's ERA normalized across the league and accounting for factors like ballpark and opponents
- ERA+=League average ERA, adjusted for ballpark factorsx100/ERA
- Example
  - Greg Maddux 1995 Braves
  - (4.18 x 1.015) x100/1.63 = 260
- Best single season for Twins
  - Johann Santana 182

Rating	ERA+
Excellent	160
Great	130
Above Average	120
Average	100
Below Average	90
Poor	80
Awful	70

Other Pitching Metrics

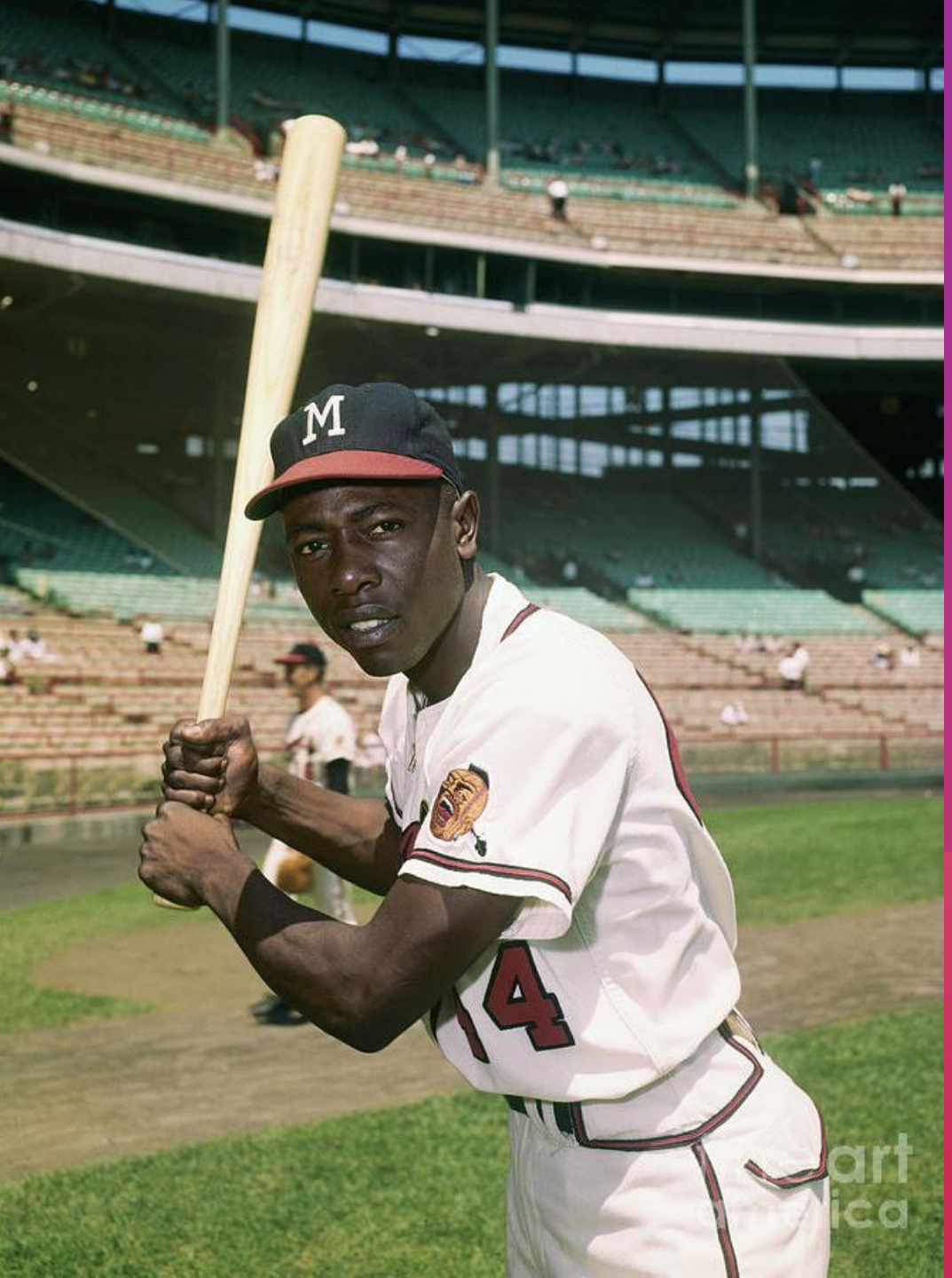
- Fly Ball rate (%), Ground Ball rate (%)
- Strike out rate (%K), Walk rate (BB%)
- K/BB ratio
- Inherited Runners Scored rate (IRS%)
- Relief Pitcher Specific
  - Saves
  - Holds



ADVANCED FIELDING METRICS

Advanced Fielding Metrics

- Traditional metrics
  - Errors
  - Assists
  - Put outs
  - Don’t take into consideration range of the fielder, ball park dimensions etc
- Gold glove awards
  - Objective
  - Not necessarily based on data
  - Two of the historically best defensive infielders elected to the HOF based on fielding prowess
    - Bill Mazeroski
    - Ozzie Smith
- Defensive Runs Saved (DRS)
  - Total runs saved compared to an average fielder
  - Wilber Abreu (Bos) DRS=28
  - Bryce Turang (MIL) DRS=23
- Defensive Efficiency Ratio (DER)
  - Team rate of turning batted balls into outs
  - DER= 1-(H+ROE-HR/PA-BB-KO-HBP-HR)



WINS ABOVE REPLACEMENT(WAR)

MEASURE OF A PLAYER'S VALUE IN ALL FACETS OF THE GAME, DETERMINING HOW MANY MORE WINS WHEN THEY ARE IN THE GAME VERSUS A REPLACEMENT

WAR

WAR for Position Players

- (The number of runs above average a player is worth based on batting, baserunning and fielding+adjustment for position+ adjustment for league+the number of runs provided by a replacement)/Runs per Win

https://www.samford.edu/sports-analytics/fans/2023/Sabermetrics-101-Understanding-the-Calculation-of-WAR

WAR for Pitchers

- Runs allowed per 9 innings adjusted for league and ballpark
- Using league averages determines how many wins the pitcher provides

WAR Rankings

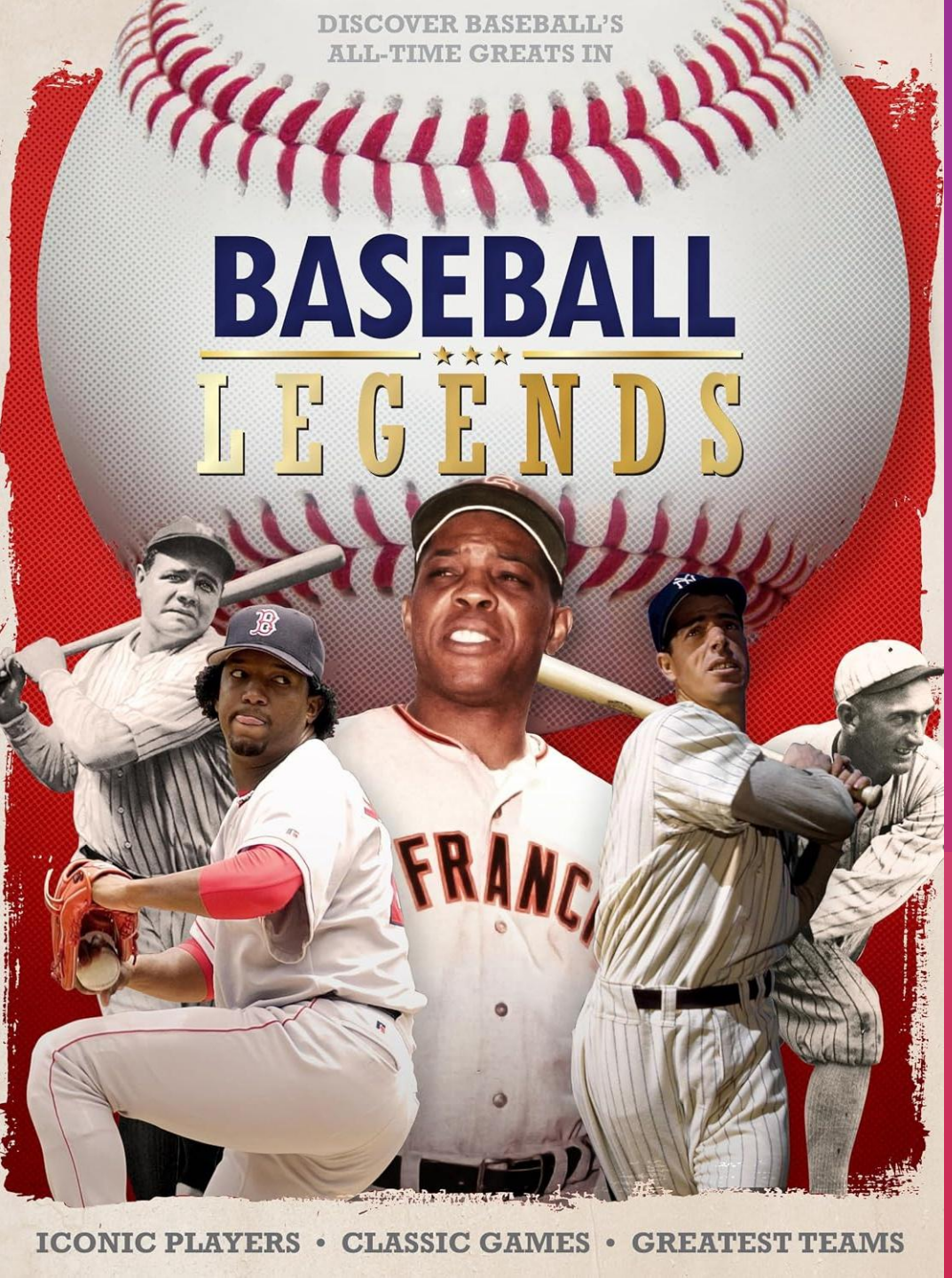
WAR Rankings

Rank	WAR
MVP material	8 or higher
Superstar	6-8
All-Star	4-6
Solid player	2-4
Role player	1-2
Bench	0-1
Back to the Minors	0 or below

MLB Leaders in WAR

Player	WAR
Crow-Armstrong	9.8
Sanchez	8.1
Turang	7.3
McGonigle	7.1
Schlittler	7
Alvarez	6.8

Twins Leaders Keaschall 3.0; Taj Bradley 2.8
Brooks Lee 2.8, Byron Buxton 2.6, Ryan Jeffers 2.6



ADVANCED METRICS AND PLAYER COMPARISONS

RANKING PLAYERS OF DIFFERENT ERAS, MILESTONES FOR HOF, BEST TWINS PLAYERS OF ALL-TIME

Comparing Players from Different Eras

- How do you compare player performance across multiple eras in baseball
  - Statistics
  - Stories/Lore
- What are the milestones that define a Hall of Fame Player
  - Hitters-.300+ Batting Avg; 3,000 Hits, 500 HRs
  - Pitchers-300 Wins, 3,000 Strikeouts, 500 Saves
  - What will be the new milestones



Example-Recent Sports Illustrated Ranking of the Top 25 Players of All-Time

- https://www.si.com/mlb/ranking-the-25-greatest-baseball-players-of-all-time
- Based upon traditional and advanced metrics

SI Top 25 Players of All-Time

Rank	Player	Peak Years	Notable Metric
25	Pedro Martinez	1994-2005	219 W, 2.93 ERA, 3,154 K, 81.4 bWAR
24	Tom Seaver	1969-1977	311 W, 2.86 ERA, 3,640 K, 109.9 bWAR
23	Rogers Hornsby	1917-1929	358/.434/.577, 301 HR, 2,930 H, 1,584 RBIs, 127.3 bWAR
22	Randy Johnson	1990-2002	303–166, 3.29 ERA, 4,875 SO, 4,135 1/3 IP, 101.1 bWAR
21	Frank Robinson	1960-1966	.294/.389/.537, 586 HR, 1,812 RBIs, 2,943 H, 107.2 bWAR
20	Christy Mathewson	1905-1913	373–188, 2.13 ERA, 2,507 SO, 4,788 IP, 106.5 bWAR
19	Rickey Henderson	1980-1991	279/.401/.419, 297 HR, 3,055 H, 1,406 SB, 111.2 bWAR
18	Roger Clemens	1986-2001	354–184, 3.12 ERA, 4,672 SO, 4,916 2/3 IP, 138.7 bWAR
17	Cy Young	1892-1904	511–315, 2.63 ERA, 2,803 SO, 7,356 IP, 163.6 bWAR
16	Mike Schmidt	1974-1986	67/.380/.527, 548 HR, 1,595 RBIs, 2,234 H, 106.9 bWAR
15	Mike Trout	2012-2019	.294/.406/.570, 404 HR, 1,018 RBIs, 1,754 H, 87.4 bWAR
14	Joe Dimaggio	1936-1942, 1946-1951	.325/.398/.579, 361 HR, 1,537 RBI, 2,214 H, 79.2 bWAR
13	Greg Maddux	1988-2000	355-227, 3.16 ERA, 3,371 SO, 5,008 1/3 IP
12	Ken Griffey Jr.	1989-2000	.284/.370/.538, 630 HR, 1,836 RBI, 2,781 H, 83.8 bWAR

SI Top 25 Players of All Time

Rank	Player	Peak Years	Metrics
11	Stan Musial	1942-1962	.331/.417/.559, 475 HR, 1,951 RBI, 3,630 H, 128.7 bWAR
10	Walter Johnson	1909-1924	417–279, 2.17 ERA, 3,509 SO, 5,914 1/3 innings, 164.8 bWAR
9	Lou Gehrig	1926-1938	340/.447/.632, 493 HR, 1,995 RBI, 2,721 H, 113.7 bWAR
8	Shohei Ohtani	2018-present	.282/.374/.582, 280 HR, 669 RBI, 1,050 H; 39–20, 3.00 ERA, 670 SO, 528 2/3 innings, 51.5 bWAR
7	Mickey Mantle	1952-1964	.298/.421/.557, 536 HR, 1,509 RBI, 2,415 H, 110.3 bWAR
6	Ted Williams	1939-1958	.344/.482/.634, 521 HR, 1,839 RBI, 2,654 H, 121.8 bWAR
5	Ty Cobb	1907-1925	.366/.433/.512, 117 HR, 1,944 RBI, 4,189 H, 149.7 bWAR
4	Hank Aaron	1955-1971	.305/.374/.555, 755 HR, 2,297 RBI, 3,771 H, 143.3 bWAR
3	Barry Bonds	1990-2004	.298/.444/.607, 762 HR, 1,996 RBI, 2,935 H, 162.8 bWAR
2	Willie Mays	1954-1966	.301/.384/.557, 660 HR, 1,909 RBI, 3,293 H, 156.2 bWAR
1	Babe Ruth	1919-1937	.342/.474/.690, 714 HR, 2,873 H, 2,214 RBI, 162.2 bWAR

Discussion

- Who shouldn’t be on the SI list
- Who should be on the SI list
- What criteria are you considering



HOW DO THE TWINS' HOF PLAYERS MEASURE UP

POSITION PLAYERS AND PITCHERS

Twin's HOF Position Players

- Harmon Killebrew-compared to other HOF Sluggers
  - 8 40+ HR seasons, second to only Babe Ruth
- Rod Carew-compared to other HOF second basemen
  - Ranks 3-6th in offensive categories
- Kirby Puckett-compared to other HOF outfielders
  - Considering peak years ranks 8-10th among CFs
- Joe Mauer-compared to other HOF catchers
  - Ranks 5-9th
- Tony Oliva-compared to other HOF outfielders
  - Peak years comparable to other HOF OF's like Billy Williams and Al Kaline

Twins' HOF Pitchers

- Bert Blyleven
  - 27th in W's, 5th in KO's, 9th in Shutouts
- Jim Kaat
  - 20th in Wins, 2nd in gold gloves



METRICS AND 2026 AWARD WINNERS

MVP, CY YOUNG AND ROOKIE OF THE YEAR

What do Metrics tell us about MVP for 2026

AL Candidates

- Yordan Alvarez (Hou)
  - AL batting title, second in HRs.1.027 OPS
- Junior Camerino (TB)
  - 41 HRs, 0.892 OPS
- Ben Rice (NY)
  - 39 HRs, high OBP

NL Candidates

- Pete Crow-Armstrong (Chi)
  - 45 HR, 38 SB, WAR leader
- Brice Turang (Mil)
  - 100 Rbi, top defensive player
- Shohei Ohtani (LA)
  - 30+ HR, two-way player

Cy Young 2026

AL Candidates

- Cam Schlittler (NY)
  - 1.95 ERA, 14 W
- Sonny Gray (Bos)
  - 18W, 2.72 ERA
- Drew Rasmussen (TB)
  - 16W, 2.67 ERA

NL Candidates

- Jacob Misiorowski (MIL)
  - 16W, 1.8 ERA, WHIP
- Christopher Sanchez (Phil)
  - 18W, leader in IP, 2.91 ERA
- Yoshinobu Yamamoto (LA)
  - 14 W, 2.53 ERA

Rookie of the Year 2026

AL Candidates

- Kevin McGonigle (Det)
  - .278 BA, 18 HR, WAR
- Parker Messick (Clev)
  - 14W, 200K
- Munetaka Murakami (CWS)
  - 33 HRs

NL Candidates

- Sal Stewart (Cin)
  - 33HR, 111 Rbi
- Nolan McLean (NYM)
  - 12 W, 195K
- JJ Wetherholt (StL)
  - 17 HRs